

ARJUN MEHTA

Senior Machine Learning Engineer

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SUMMARY

Machine learning engineer with 5 years of experience building production ML systems across recommendation, NLP, and computer vision. Shipped models that served over 40 million monthly active users across fintech and e-commerce platforms. Strong background in large-scale training pipelines, feature engineering, and A/B experimentation. Seeking senior ML engineer roles in the United States; will require H-1B visa sponsorship from the prospective employer. Compensation expectation in the range of INR 65-75 lakhs per annum for India-based roles, with flexibility to align with U.S. market benchmarks for sponsored positions.

WORK EXPERIENCE

Senior Machine Learning Engineer

Razorpay Software Pvt. Ltd., Bangalore, India

March 2023 - Present

- Designed and deployed a real-time fraud detection system using a gradient-boosted ensemble (XGBoost + LightGBM) over 180+ engineered features, reducing payment fraud losses by 31 percent and saving an estimated INR 84 crore annually across 12 million monthly transactions.
- Led the migration of the legacy batch scoring pipeline to a streaming architecture on Apache Flink and Kafka, cutting end-to-end inference latency from 14 seconds to under 220 milliseconds for the risk decisioning service.
- Architected the merchant lifetime value (LTV) prediction model using Temporal Fusion Transformers, integrated into the CRM tool used by 800+ relationship managers; improved upsell conversion by 19 percent in controlled A/B tests over 60 days.
- Built an internal ML platform feature store (Feast-based) serving 2.4 TB of online features with sub-50ms p99 lookup, adopted by six downstream teams and eliminating approximately 70 percent of redundant feature computation.
- Mentored four junior engineers through structured biweekly ML system design reviews; two were promoted to mid-level within 11 months.

Machine Learning Engineer

Flipkart Internet Pvt. Ltd., Bangalore, India

July 2021 - February 2023

- Built the next-best-action recommender for the post-order tracking page using a two-tower neural retrieval model followed by a cross-attention ranker, increasing repeat purchase rate by 7.4 percent and incremental GMV of INR 220 crore over 9 months.
- Co-developed Flipkart's session-based personalization service on TensorFlow Serving, handling 18,000 QPS during peak Big Billion Days traffic with a documented availability of 99.94 percent.
- Created an off-policy evaluation framework using IPS and doubly robust estimators that reduced the time-to-decision for new ranking models from 6 weeks to 9 days, saving roughly 280 engineering hours per quarter.
- Owned the cold-start problem for new seller listings by combining product attribute embeddings (from a fine-tuned MiniLM) with category-level priors, lifting first-week impressions for new SKUs by 23 percent.

Associate Machine Learning Engineer

Mu Sigma Business Solutions Pvt. Ltd., Bangalore, India

August 2019 - June 2021

- Delivered churn prediction models for two Fortune 500 telecom clients (Verizon Wireless, Deutsche Telekom) using stacked ensembles; achieved AUC of 0.87 and 0.84 respectively on holdout sets.
- Built a demand forecasting system for a U.S. grocery client using Prophet and LSTM hybrids, reducing mean absolute percentage error from 18.6 percent to 9.2 percent across 1,400 store-SKU combinations.
- Authored an internal NLP toolkit (tokenization, NER wrappers, and weak-supervision labeling using Snorkel) adopted by 11 client teams, cutting typical text pipeline setup time by approximately 60 percent.

PROJECTS

BharatTranslate - Low-Resource Neural Machine Translation

github.com/arjunm-iitb/bharattranslate

- Built a 250M-parameter Transformer model translating between 11 Indian language pairs (Hindi, Tamil, Bengali, Telugu, Marathi, Malayalam, Kannada, Gujarati, Punjabi, Odia, Assamese), training on a curated 18M sentence-parallel corpus augmented with back-translation.
- Achieved a BLEU score of 28.4 on Hindi-Tamil (compared to 14.1 from the IndicTrans baseline) and won the Best Open-Source Project award at ML India Annual Summit 2022.

CardioRisk - ECG-Based Cardiovascular Risk Stratification

- Developed a 1D ResNet-50 model trained on 1.2 million 12-lead ECG recordings (PTB-XL and a private hospital dataset) to predict 5-year major adverse cardiac events, reaching an AUROC of 0.91.
- Deployed as a clinician-facing dashboard at a partner hospital in Mumbai; flagged 312 high-risk patients in a 3-month pilot, of whom 274 were independently confirmed positive on follow-up angiography.

EDUCATION

Bachelor of Technology (B.Tech.), Computer Science and Engineering

Indian Institute of Technology Bombay, Mumbai, India

2015 - 2019

CGPA: 9.18 / 10.0

Relevant Coursework: Machine Learning, Deep Learning, Optimization, Probabilistic Graphical Models, Distributed Systems, Information Retrieval.

Capstone: "Sparse Mixture-of-Experts for Multilingual Document Classification" (Advisor: Prof. Preethi Jyothi), published as a workshop paper at EMNLP 2020.

SKILLS

Programming: Python, Scala, Go, SQL, Bash, C++

ML Frameworks: PyTorch, TensorFlow, JAX, scikit-learn, XGBoost, LightGBM, CatBoost

NLP / CV: Hugging Face Transformers, spaCy, OpenCV, Sentence-Transformers, FAISS

Data & Infrastructure: Apache Spark, Kafka, Flink, Airflow, dbt, Ray, MLflow, Kubeflow, Feast, Redis, PostgreSQL, BigQuery, Snowflake

Cloud: AWS (S3, SageMaker, EKS, Lambda), GCP (Vertex AI, GKE), Kubernetes, Docker, Terraform

Practices: A/B testing, causal inference, offline/online evaluation, feature stores, model governance, MLOps, cost optimization, technical mentoring

ADDITIONAL

- Publications: "Sparse Routing for Low-Resource Code-Mixed Hinglish Classification," EMNLP 2020 Workshop on NLP for Similar Languages and Code-Switching.
- Certifications: AWS Certified Machine Learning - Specialty (2023), Deep Learning Specialization, deeplearning.ai

(2020).

- Languages: English (fluent), Hindi (native), Marathi (conversational).
- Visa status: Indian citizen; requires H-1B sponsorship for U.S.-based employment. Already in possession of a valid U.S. B1/B2 visa (expiring March 2028) and have previously traveled to the U.S. on business four times for client engagements.